TECHNICAL SUPPLEMENT 06 · ENGLISH

Weight Sensitivity and Sampling Uncertainty

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Different sources of variation

Weight sensitivity changes the stated preference vector while holding the observed sample fixed. Within-cell bootstrap varies the sampled attempts and does not verify pairing or recover unobserved real-world variability. The archived outputs contain 500 weight vectors and 1000 independent within-cell bootstrap resamples per scenario/pooled set. This document summarizes those recorded outputs; rebuilding this PDF does not rerun their stochastic generation.

POOLED

Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
M0	0.685	[0.669, 0.703]	7.000	0.000	1.000	0.112
M1	0.157	[0.135, 0.175]	9.000	0.000	0.000	0.000
M2	0.160	[0.139, 0.179]	8.000	0.000	0.000	0.000
M3	0.788	[0.777, 0.798]	5.932	0.000	0.000	0.000
M4	0.843	[0.838, 0.847]	3.009	0.000	0.019	0.000
B1	0.812	[0.790, 0.836]	4.559	0.000	0.000	0.000
B2	0.814	[0.784, 0.836]	4.500	0.000	0.305	0.000
B3	0.921	[0.914, 0.927]	1.026	0.974	1.000	0.674
B4	0.895	[0.868, 0.922]	1.974	0.026	1.000	0.214

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Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
M0	0.605	[0.559, 0.659]	7.000	0.000	1.000	0.000
M1	0.182	[0.115, 0.237]	8.385	0.000	0.000	0.000
M2	0.182	[0.114, 0.237]	8.615	0.000	0.000	0.000
M3	0.838	[0.816, 0.857]	5.997	0.000	0.000	0.000
M4	0.882	[0.872, 0.889]	4.259	0.000	0.000	0.000
B1	0.908	[0.892, 0.923]	2.903	0.000	0.000	0.000
B2	0.875	[0.859, 0.892]	4.735	0.000	0.673	0.028
B3	0.919	[0.911, 0.926]	2.106	0.000	0.879	0.122
B4	0.975	[0.972, 0.978]	1.000	1.000	1.000	0.850

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Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
M0	0.814	[0.813, 0.815]	7.000	0.000	1.000	0.174
M1	0.008	[0.006, 0.011]	8.341	0.000	0.000	0.000
M2	0.008	[0.006, 0.010]	8.659	0.000	0.000	0.000
M3	0.891	[0.888, 0.895]	3.219	0.000	0.000	0.000
M4	0.884	[0.882, 0.886]	4.436	0.000	0.000	0.000
B1	0.882	[0.856, 0.904]	4.346	0.000	0.000	0.000
B2	0.844	[0.842, 0.845]	5.999	0.000	1.000	0.000

Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
B3	0.943	[0.942, 0.945]	1.772	0.228	1.000	0.486
B4	0.946	[0.940, 0.952]	1.228	0.772	1.000	0.340

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Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
M0	0.590	[0.588, 0.592]	7.000	0.000	1.000	0.000
M1	0.230	[0.228, 0.232]	9.000	0.000	0.000	0.000
M2	0.243	[0.240, 0.245]	8.000	0.000	0.000	0.000
M3	0.877	[0.865, 0.888]	4.792	0.000	0.000	0.000
M4	0.893	[0.890, 0.895]	3.560	0.000	0.000	0.000
B1	0.893	[0.850, 0.925]	3.656	0.000	0.000	0.000
B2	0.830	[0.806, 0.857]	5.992	0.000	0.000	0.000
B3	0.931	[0.930, 0.932]	2.000	0.000	1.000	0.302
B4	0.970	[0.969, 0.970]	1.000	1.000	1.000	0.698

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Method	C mean	C 95% range	Mean rank	Top1 boot	P non-dom	Top1 weights
M0	0.729	[0.695, 0.772]	2.520	0.000	1.000	0.398
M1	0.183	[0.125, 0.221]	8.100	0.000	0.000	0.000
M2	0.181	[0.123, 0.220]	8.900	0.000	0.000	0.000
M3	0.558	[0.527, 0.589]	6.442	0.000	0.014	0.000
M4	0.684	[0.663, 0.703]	4.127	0.000	0.740	0.000
B1	0.555	[0.488, 0.631]	6.518	0.000	0.017	0.000
B2	0.699	[0.570, 0.777]	3.377	0.000	0.399	0.000
B3	0.870	[0.833, 0.897]	1.000	1.000	1.000	0.598
B4	0.681	[0.597, 0.778]	4.016	0.000	1.000	0.004

Criterion-removal ranks / Rangos al retirar un criterio

Preference	M0	M1	M2	M3	M4	B1	B2	B3	B4
('1', 'MissionClearance')	7	8	9	6	5	3	4	2	1
('1', 'MissionExposure')	7	9	8	6	5	4	3	2	1
('2', 'MissionClearance')	2	8	9	7	5	6	3	1	4
('2', 'MissionExposure')	7	8	9	3	4	5	6	1	2
('3', 'MissionClearance')	7	9	8	6	4	5	3	2	1
('3', 'MissionExposure')	7	8	9	5	4	3	6	2	1
('4', 'MissionClearance')	3	8	9	7	4	6	2	1	5
('4', 'MissionExposure')	2	9	8	7	4	6	3	1	5
('POOLED', 'MissionClearance')	5	9	8	7	4	6	2	1	3

Preference	M0	M1	M2	M3	M4	B1	B2	B3	B4
('POOLED', 'MissionExposure')	7	9	8	6	3	5	4	1	2

Scenario-preference ranks / Rangos según preferencias de escenario

Preference	M0	M1	M2	M3	M4	B1	B2	B3	B4
0	7	9	8	6	3	5	4	1	2
1	7	9	8	6	4	3	5	2	1
2	7	9	8	5	3	4	6	1	2
3	7	9	8	5	3	4	6	2	1
4	5	9	8	7	3	6	4	1	2

Top1 weights is the fraction with reported rank 1; fractionalTop1 is the archived bootstrap tie-aware first-rank summary. Marginal 95% closeness ranges are not simultaneous confidence intervals for all method rankings. Preference profiles and scenario weights are author-defined sensitivity settings, not expert-elicited task requirements.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.